

Contextual Bandits

X = contexts / states

P_X - distribution over X

$\mathcal{A}$ = actions / arms

$p(r|x, a) \quad \forall a \in \mathcal{A}$

π - ?

$$R_T(\pi) = \mathbb{E}_{x_t \sim P_X, a_t \sim \pi(\cdot|x_t), r_t \sim p(r|x_t, a_t)} \left[\sum_{t=1}^T r(x_t, a_t^*) - r(x_t, a_t) \right]$$

$$a_t^* = \arg\max_a \mathbb{E}[r(x_t, a)]$$

$$\pi^* = \arg\max_{\pi} R_T(\pi)$$

$$x_0 \xrightarrow{\pi_0} a_0 \rightarrow r_0$$

$$x_1 \xrightarrow{\pi_1} a_1 \rightarrow r_1$$

$$\text{UCB} \quad a_k | x_k = \arg\max_{a_k} \left[\hat{Q}_{k-1}(x_k, a_k) + c \sqrt{\frac{\log k}{n_k(a_k, x_k)}} \right]$$

$$n_k(a_k, x_k) = \#(a, x) = (a_k, x_k)$$

$$\hat{Q}_k(x_k, a_k) = \frac{n_k(a_k, x_k)}{n_k(x_k)}$$

Contextual Linear Bandits

$$\mathbb{E}[r(x_t, a_t)] = x_t^T \theta^* = x_t^T \theta_a$$

LinUCB

$$a_{k|X_k} = \arg\max_a X_t^T \theta_{a,t} + \alpha \sqrt{X_t^T A_{a,t}^{-1} X_t}$$

$$\theta_{a,t} = A_{a,t}^{-1} b_{a,t}$$

$$A_{a,t} = I + \sum_{t' | a_{t'} = a} X_{t'} X_{t'}^T = I + X_a^T X_a$$

$$b_{a,t} = \sum_{t' | a_{t'} = a} X_{t'}^T r_{a,t'}$$

$$(x_1, a_1, r_1), (x_2, a_2, r_2), \dots$$

$$r_i \approx x_i^T \theta_{a_i}, \quad X_a = \begin{bmatrix} x_{1,a} \\ x_{2,a} \\ \dots \end{bmatrix}$$

for $a \in \mathcal{A}$:

$$R_a = X_a \theta_a$$

↑
lin. regression

$$\min_{\theta_a} \sum_i (r_i - x_i^T \theta_a)^2 + \lambda \cdot \|\theta\|^2$$

$$\hat{\theta}_a = \underbrace{(X_a^T X_a + \lambda I)^{-1}}_{A_a^{-1}} \underbrace{X_a^T R_a}_{b_a}$$

$$X_t^T A_{a,t}^{-1} X_t = X^T A_a^{-1} X$$

$A \approx$ covariance matrix

$$x \sim p_x$$

$$x^T A_a^{-1} x - \text{uncertainty}, \quad r = x^T \theta^* + \varepsilon$$

$$\text{upper bound} = x^T \hat{\theta} + \alpha \sqrt{x^T A^{-1} x}$$

$$\alpha = 1,96 \text{ bzw. } 0,95$$